Can Community Cultural Wealth Maintain STEM Career Aspirations of Undergraduate Women of Color? Examining HERI Data

Abstract

Women of color (WOC) remain underrepresented in the STEM workforce due to systemic racism and sexism. Using Yosso's (2005) anti-deficit Community Cultural Wealth framework, this study quantitatively explored factors associated with STEM career aspiration persistence among senior WOC undergraduates who entered college with initial STEM career aspirations. Drawing on longitudinal data from the Higher Education Research Institute's (HERI) 2015 Freshman Survey (TFS) and 2019 College Senior Survey (CSS), we analyzed a sample of 1,460 WOC graduating seniors across 42 institutions using logistic regression. Results revealed that aspirational and navigational capital played pivotal roles in maintaining STEM career aspirations. Specifically, WOC who expressed strong desires to make scientific contributions were significantly more likely to maintain STEM career aspirations. Additionally, WOC who participated in undergraduate research and STEM support programs had significantly higher likelihoods of reporting STEM career plans at the end of their senior year. By contrast, WOC with strong resistant capital, particularly those committed to social change and community leadership, were less likely to sustain their interest in pursuing STEM careers. We conclude by discussing several recommendations for practice and future research.

Keywords: Women of color, STEM career aspirations, STEM persistence, community cultural wealth, higher education

Introduction

Historically, science, technology, engineering, and mathematics (STEM) fields have been predominantly shaped by cultural norms and values reflecting White, male, middle-class experiences, emphasizing objectivity, individualism, competition, and meritocracy (Miles et al., 2022). These prevailing cultural norms have substantially contributed to systemic inequities and systematic oppression within higher education contexts, suppressing the success of marginalized demographic groups in STEM fields (Costello et al., 2023; Denaro et al., 2022; Sax & Newhouse, 2018). Recent data highlight the persistent underrepresentation of women of color (WOC) in science and engineering (S&E) occupations within the U.S. In 2021, Asian women represented 6% of S&E occupations, followed by Hispanic or Latina women (5%), Black or African American women (4%), multiracial women (1%), and American Indian or Alaska Native (AIAN) and Native Hawaiian or Pacific Islander (NHPI) women, each accounting for merely 0.1%. By contrast, White women occupied nearly one-third (31%) of S&E jobs (NCSES, 2023). These disparities highlight the "double bind" faced by WOC, especially Black, Latina, and Native American women, who simultaneously experience sexism and racism, reinforcing assumptions about who can and should pursue STEM careers (Ong et al., 2011).

Rather than following a linear "pipeline" into the STEM workforce, students with initial STEM career aspirations find numerous entry points along their pathways toward careers both within and outside their field of study (Fry et al., 2021). These non-linear career trends are particularly pronounced among WOC, representing a significant challenge in diversifying the STEM workforces (Park et al., 2022). Barriers restricting entry for WOC into the STEM workforce include discrimination, inequitable access to resources, and exclusionary practices in both higher education and industry (Milkman et al., 2015; McKee & Delgado, 2020). For instance, WOC are frequently overlooked for leadership roles, experience microaggressions from supervisors and colleagues, and find themselves excluded from professional networks essential for career advancement (Corneille et al., 2019; Kim & Meister, 2023; Miles et al., 2022). These barriers collectively create a persistent “chilly climate” in academic and professional STEM contexts, discouraging WOC from pursuing or persisting in these fields (Jorstad et al., 2017; Simon et al., 2017). While targeted institutional support, such as mentorship, science clubs, and undergraduate research experiences, can mitigate some disparities due to historical and systemic racism (Eagan et al., 2024; Estrada et al., 2018; Ong et al., 2018; Sausner et al., 2024), research also reveals that traditional mentorship approaches may not always benefit women of color, as poorly matched mentoring relationships or mentors who lack cultural competency can inadvertently reinforce existing barriers and negatively impact career aspirations among women of color in STEM fields (Limeri et al., 2019; Park et al., 2022).

Aspirations play an important role in shaping students’ educational and career trajectories in STEM. They manifest differently depending on a student’s race, gender, and institutional context (Espinosa, 2011; Chang et al., 2014; Xu, 2016; Lehman et al., 2023). Women are more sensitive to sociopsychological influences from significant others in the development of their STEM aspirations (Xu, 2016). Institutional context also matters, as women attending more selective institutions are more likely to aspire to pursue graduate degrees in STEM fields (Xu, 2016). Despite this evidence much of the existing research on aspiration and motivation in STEM careers draws from social cognitive career theory and expectancy-value theory (e.g., Amalina et al., 2025; Fredricks et al., 2025), which, while valuable, may not adequately capture the community cultural influences that shape students’ experiences.

Recent scholarship has challenged common deficit-informed narratives that emphasize the perceived lack of resources and opportunities available to WOC (Denton et al., 2020; Valencia, 2010; Yap et al., 2024). Yosso's (2005) Community Cultural Wealth (CCW) framework provides researchers with an alternative lens to the cultural capital perspectives perpetuated by (Bourdieu & Passeron, 1990). CCW emphasizes the inherent strengths that underrepresented groups (URGs) bring with them to college. Previous research on CCW has predominantly concentrated on qualitative methodologies within higher education contexts (Connors, 2023; Rodriguez et al., 2023; Slack et al., 2024), with limited quantitative exploration of CCW's impact (Hiramori et al., 2024; Nhien, 2024; Sablan, 2019).

Drawing on the CCW framework, this study aims to contribute to the body of research exploring the persistence of WOC's STEM career aspirations and the influence of four of the six forms of CCW: aspirational, social, navigational, and resistant capital, using a quantitative approach. Additionally, this study seeks to provide actionable strategies and policy recommendations to sustain WOC students' career aspirations in STEM that leverage their inherent strengths. By analyzing longitudinal data from the Higher Education Research Institute's (HERI) 2015 Freshman Survey (TFS) and 2019 College Senior Survey (CSS), this study addresses the following research questions:

1. What demographic and family background characteristics distinguish women of color who continue to pursue STEM career aspirations throughout college from those who express interest in other career paths by their senior year?
2. Controlling for demographic and family background characteristics, to what extent are different forms of community cultural wealth (CCW) associated with the sustained pursuit of STEM career aspirations among senior women of color?

Exploring the Relevance of Community Cultural Wealth to STEM Career Aspirations

As an extension of Critical Race Theory, Yosso (2005) conceptualized CCW as “an array of knowledge, skills, abilities, and contacts possessed and utilized by Community of Color to survive and resist macro and micro-forms of oppression” (p. 77). Yosso (2005) reframes Bourdieu’s conception of cultural capital to recognize the cultural assets that historically marginalized students bring with them to higher education spaces. Her framework challenges researchers and practitioners to interrogate whose culture has capital, or is validated, within traditionally White educational contexts. These unique cultural resources nurtured by families and communities within systematically marginalized communities equip students with critical skills to persevere in predominantly White academic settings. Yosso (2005) describes six interconnected dimensions of CCW that students of color possess and develop during their education: aspirational, linguistic, familial, social, navigational, and resistant capital. In this study, we use Yosso’s (2005) CCW framework to explore how the talents, strengths, and experiences that WOC students bring to college contribute to their development and sustained pursuit of STEM career aspirations.

The Role of Familial Support and Aspirational Capital in Fostering Long-Term Plans

Students’ career aspirations often develop in relation to familial influences. Yosso (2005) describes familial capital as the social and personal connections resulting from extended familial and community-based networks. Familial influences, such as having parents with careers in STEM, particularly those holding advanced degrees, significantly correlate with students’ aspirations to pursue STEM-related careers and graduate degrees (Lehman et al., 2023; Sax et al., 2018; Xu, 2016). Students can acquire early technical skills and receive encouragement from family members to pursue STEM fields, which can have notable benefits, particularly among

Latina students, as they report greater levels of confidence associated with increased emotional support from family (Rodriguez et al., 2023).

Support from family can also contribute to students' aspirational capital, which Yosso (2005) describes as an individual's ability to maintain and sustain their lifelong goals, particularly in the face of adversity. Aspirational capital manifests as students articulate their long-term career objectives and personal values, such as addressing social and environmental justice (Connors, 2023) or advancing their STEM careers through self-driven persistence (Sausner et al., 2024). Aspirational capital may serve as a protective factor for Latina/o/x students who perceive STEM content as boring or irrelevant to their personal interests and goals. Moreover, WOC tend to have stronger altruistic components in their aspirational capital, such as plans to establish health clinics in underserved communities. These altruistic aspirations nourish their development of science identity and STEM pathways (Carlone & Johnson, 2007).

Networks and Determination: Considering Influences from Navigational, Social, and Resistant Capital

After students enter college, their navigational, social, and resistant capital inform their development of their longer-term career objectives. Yosso (2005) explains navigational capital as the strategies and skills individuals use to navigate social institutions and interactions, such as actively participating in targeted programs promoting STEM success and identifying research opportunities (Rincón & Rodriguez, 2021). Such involvement is associated with increased persistence in STEM fields (Chang et al., 2014) and higher aspirations for STEM graduate studies (Eagan et al., 2013).

Moreover, engagement in undergraduate research opportunities further supports students by strengthening their science identity, offering hands-on experience, and fostering meaningful faculty mentorship, significantly reinforcing their aspirations for STEM graduate studies and careers (Eagan et al., 2024; Hernandez et al., 2018; Wilson et al., 2018). Additionally, participation in undergraduate research exposes students to disciplinary norms, graduate-level expectations, and insider knowledge that can facilitate their pathways into either graduate school and/or the STEM workforce (Eagan et al., 2013; Hernandez et al., 2018). Recent empirical studies also show that undergraduate internship opportunities help students of color develop essential skills for navigating professional spaces, build meaningful professional networks, and acquire working experiences necessary for STEM career advancement (Elkhider et al., 2025; Rosenzweig et al., 2024).

Relatedly, social capital refers to how students' connections with peers and other institutional agents in their social network collectively contribute to their development (Yosso, 2005). Peer networks facilitate discussions about financing education and navigating institutional challenges, collectively offering vital support and access to academic opportunities (Slack et al., 2024). These networks and activities foster a sense of belonging, provide mutual support, and offer access to academic and career resources, potentially increasing their intentions to pursue STEM careers (Chang et al., 2014; Rincón & Rodriguez, 2021). Students also leverage their social capital through faculty-student mentorship, which offers technical guidance, emotional support, and access to resources and networks that enhance academic and research opportunities, empowering them to pursue STEM careers, especially among URGs students (Thompson & Jensen-Ryan, 2018). Moreover, internships and other STEM-related work experiences during college may also serve as social capital, as students establish professional relationships and gain exposure to real workplace environments, which can further sustain their interest in STEM careers (Garibay et al., 2013).

Finally, resistant capital refers to students' knowledge and skills developed through confronting and seeking to address social inequality (Yosso, 2005). It often manifests in their advocacy efforts to disrupt stereotypes and promote equity within their academic environments, as exemplified by those who confront inequities in classroom or campus settings (Connors, 2023; Rincón & Rodriguez, 2021). In particular, as students see themselves and their communities better reflected in their STEM curriculum and in the research topics they pursue, they may be more inclined to persist in their STEM majors and see STEM careers as a viable pathway (Garibay, 2018).

The empirical and conceptual pieces described above demonstrate that CCW provides an asset-based perspective for understanding how students of color navigate and persist in STEM fields by leveraging their personal and community assets within the higher education context. In this study, we extend research on the influence of CCW on WOC's STEM career aspiration persistence and provide a more critical reflection based on this transformative framework.

Applying CCW in Quantitative Studies

Several studies have quantitatively explored the connection between elements of the college experience and the cognitive and psychosocial outcomes of STEM students using a CCW lens (Hiramori et al., 2024; Nhien, 2024; Sablan, 2019). Sablan (2019) quantitatively measured CCW through survey scales assessing aspirational, familial, navigational, and resistant capital, highlighting how quantitative measurement can deepen understanding of each construct. Building on Sablan's (2019) work, Nhien's (2024) study provides a notable example of applying quantitative methods to evaluate CCW influence on the science self-efficacy of Southeast Asian students. Specifically, Nhien (2024) measured aspirational capital through leadership, social goals, and incoming first-year students' plans to pursue science-related careers. He operationalized familial capital through interactions with family, feelings about family support, and the impact of family responsibilities on schoolwork. This study found that among the different forms of CCW, resistant capital, specifically students' ability to adjust to the academic demands of college, can positively predict science self-efficacy, while social and navigational capital were not significant predictors. Additionally, a recent study highlights the interconnectedness of different forms of CCW, showing how dimensions like aspirational, familial, and navigational capital overlap and interact dynamically through sub-dimensions such as aspirational familial capital (Hiramori et al., 2024).

Building on existing qualitative and quantitative research and drawing from Nhien's (2024) work, we consider how participation in academic clubs, receiving mentorship from faculty, and engaging in undergraduate research experiences extend traditional notions of what constitutes navigational, social, and resistant capital. We also explore how students' commitment to social justice may provide them with additional tools to sustain their interests in pursuing STEM-related careers. We describe these and other relevant covariates in the next section.

Methods

Data Source and Sample

The data for this study were drawn from two longitudinal surveys administered by the Higher Education Research Institute (HERI): 2015 Freshman Survey (TFS) and 2019 College Senior Survey (CSS). The TFS provides a baseline assessment of incoming first-year college students, capturing information on demographics, familial background, pre-college experiences, educational and career aspirations, and psychosocial characteristics. The College Senior Survey (CSS) collects data on students' academic and co-curricular experiences, career concerns, and post-graduation plans. For this study, the sample was limited to WOC graduating seniors who entered college with an intention to pursue a STEM career and responded to both surveys. We selected graduating seniors as the study focus because they are at a pivotal decision-making stage, as they prepare to transition from undergraduate to post-baccalaureate pathways, making it an ideal moment to examine their career aspirations.

The missing data rate for variables ranged from 1% to 12%. For continuous variables, missing data were imputed using expectation-maximization (EM) algorithms after removing outliers. For categorical variables, including senior year STEM career aspirations, WOC freshman year major fields, as well as demographic variables such as race, citizenship status, first-generation status, and parental career status, observations with missing data were excluded through listwise deletion.

Our final analytic sample consisted of 1,460 WOC seniors who entered college with an intention to pursue a STEM career across 42 institutions. The sample included 43% Asian, 18% Hispanic, 10% Black, 27% identifying as multiracial, and 2% identifying as Native American or al/ethnic backgrounds. Within the Asian subgroup, approximately 45% identified as East Asian, 30% as Southeast Asian, 20% as South Asian, and 5% as Other Asian. Although Asian women as an aggregate group are not underrepresented in STEM by numeric representation, we included them in our study of WOC, as their experiences are shaped by racialized and gendered stereotypes, including the model minority myth of academic passivity and presumed excellence, that historically have obscured within-group diversity and unique barriers to belonging and advancement (Miles et al., 2022). Table 1 displays descriptive statistics of the variables used in this study.

Dependent Variable

Our key dependent variable, derived from the 2019 CSS, represents the aspirations to pursue STEM careers among senior WOC. The dichotomous variable has a value of 1 indicating an aspiration for a STEM career or 0 indicating a non-STEM career aspiration when they prepared for graduation. WOC who skipped the question or answered “undecided” were excluded from the study. The classification of STEM careers includes a diverse range of professions, including but not limited to research scientists, computer programmers, engineers, and home health workers. A full list of STEM careers in both TFS and CSS is provided in Appendix A.

Independent Variables

Our independent variables, drawn from both 2015 TFS and 2019 CSS surveys, were grouped into three blocks: (1) personal inputs and background characteristics, (2) aspirational and resistant capital from TFS, with a focus on psychological motivations and personal values, and (3) navigational and resistant capital from CSS, with an emphasis on students’ college experiences.

Personal Inputs and Background Characteristics

The model incorporated several background characteristics, including race/ethnicity, STEM major fields, first-generation college status, citizenship status, and parental occupation. Due to the small number of Native American respondents (n = 2), this group was combined with the ‘Other’ racial category to preserve statistical stability. Given the study’s focus on WOC as a collective group, we employed effect coding (Mayhew & Simonoff, 2015) rather than dummy coding to avoid designating any single racial group as the reference category. This approach allows the estimated differences for each racial group to be interpreted relative to the grand mean of the full sample, aligning with our equity-centered analytic framework.

To capture disciplinary variation, we included freshman year major subfields as a set of dummy variables reflecting the unique academic cultures across different disciplinary fields (Sax & Newhouse, 2018). The categories were Non-STEM, Math and Computer Science, Engineering, Physical Science, and Health Professions, with Biological and Life Sciences as the reference group. The full list of classifications is presented in Appendix B. To account for the possibility that students with higher academic achievement are more likely to pursue STEM careers (Rosenzweig et al., 2024; Xu, 2016), we included college GPA from the CSS, measured on a scale from 1 (D) to 8 (A or A+). For the logistic regression analysis, college GPA was standardized to have a mean of 0 and a standard deviation of 1.

Additionally, we controlled for several key background characteristics based on students' responses to the 2015 TFS. First-generation college status was included as a binary variable to capture whether students were the first in their families to attend college. We included dummy variables representing citizenship status, comparing international students, permanent residents, U.S. citizens to those who marked "none of the above," which served as the reference group. To capture the influence of family background on career aspirations (Xu, 2016), we included a binary indicator of parental occupation in STEM, reflecting whether at least one parent worked in STEM fields. Furthermore, to address variation in institutional context, we included institutional control (public vs. private) and institutional selectivity, measured by the average of the 25th and 75th percentile SAT scores, as covariates.

Measures of Aspirational and Resistant Capital

Our primary independent variables of interest are the CCW forms of capital. Aspirational and resistant capital, derived from the 2015 TFS, capture WOC's initial psychological motivations and personal values toward future careers they brought with them when entering college. All non-binary CCW measures were standardized as z-scores with a mean of 0 and a standard deviation of 1 prior to inclusion in the logistic regression analysis. This enables regression coefficients to be interpreted as the estimated change in the outcome associated with a one-standard deviation increase in the predictor, aligning the interpretation with standardized effect sizes. In this study, we measured aspirational capital using three ordinal items from the 2015 TFS, each rated on a four-point Likert scale. These items captured students' self-reported personal goals at the time of entering college, including becoming an authority in their field, making a theoretical contribution to science, and receiving professional recognition from colleagues. To preserve the nuance of each aspirational goal, we analyzed the items separately rather than combining them into a single scale.

We measured resistant capital using three indicators derived from 2015 TFS: students' freshman year commitment to community leadership and advocacy for social change (i.e., HERI's TFS social agency score, with an original mean of 50 and standard deviation of 10); their freshman year intentions to participate in student protests or demonstrations; and HERI's TFS civic engagement latent measure, which captures the extent to which WOC were motivated and involved in civic, electoral, and political activities (Sharkness et al., 2010). These measures collectively highlight WOC's initial commitment to and engagement in various forms of resistance against oppressive systems within higher education and society.

Measures of Navigational and Social Capital

Navigational and social capital, derived from the 2019 CSS, capture WOC's college experiences and their ability to maneuver through institutional structures while building meaningful relationships and support networks that facilitate academic and career success. Several prior studies informed our operationalization of the CCW (e.g., Hiramori et al., 2024;

Nhien, 2024; Sablan, 2019). We operationalized navigational capital using three binary indicators from 2019 CSS, reflecting students' engagement with campus-based programs and activities that support academic success, skill-building, and career preparation in STEM fields. First, we included a dichotomous variable indicating whether students had participated in undergraduate research, as prior research has shown that such experiences can foster academic identity, facilitate faculty mentorship, and enhance STEM career aspirations (Fredricks et al., 2025; Park et al., 2022). We also accounted for participation in federally funded campus-based STEM support programs, such as BUILD, MARC, HHMI, or S-STEM[1]. These programs are designed to support URGs by giving them insight into disciplinary norms and graduate-level expectations through activities connected to mentored research experiences and academic enrichment (Cuellar & Gonzalez, 2021; Eagan et al., 2024). Additionally, a binary variable indicating participation in internship programs was included as an indicator of navigational capital, as these experiences can provide valuable networking opportunities, expose students to professional environments, and develop practical skills relevant to their chosen STEM fields (Rosenzweig et al., 2024; Thompson & Jensen-Ryan, 2018).

Similarly, we measured social capital using three items from the 2019 CSS. The first indicator was faculty mentorship, a factor constructed by HERI, which represents the frequency students received mentorship from faculty. The second indicator was a single CSS item measuring how often students met with a career advisor, which was measured on a three-point scale ranging from "never" to "frequently." The last indicator was a binary variable, indicating involvement in pre-professional or departmental STEM clubs, as these clubs can facilitate

[1] BUILD (Building Infrastructure Leading to Diversity), MARC (Maximizing Access to Research Careers), HHMI (Howard Hughes Medical Institute), and S-STEM (Scholarships in Science, Technology, Engineering, and Mathematics).

students’ access to academic resources, connections with faculty and professionals, and early exposure to STEM careers (Chang et al., 2014).

Analytic Approach

To explore the characteristics of our sample and the extent to which different forms of CCW are associated with STEM career aspirations among graduating WOC who entered college with a STEM career aspiration, we conducted a two-phase analytic approach. Although participants were enrolled across 42 higher education institutions, the intraclass correlation coefficients (ICCs) for the outcome variable were below 0.05, suggesting small between-institution variance (Raudenbush & Bryk, 2002). Additionally, robust standard errors clustered by institution produced results nearly identical to those generated by model-based standard errors. Based on this, we proceeded with single-level analyses.

The first phase of analysis involved descriptive statistics and several cross-tabulations to identify patterns in STEM career aspirations across key demographic, academic, and institutional variables. Chi-square tests were used to examine whether the likelihood of pursuing a STEM career or non-STEM career varied significantly by characteristics such as race/ethnicity, freshman year major field, citizenship status, college GPA category, and familial background.

In the second phase of analysis, we applied binary logistic regression to examine how various CCW-related predictors were associated with students’ intentions to pursue STEM careers. We reported odds ratios (OR) to indicate the direction and magnitude of each association: values above 1 suggest a positive relationship with the dependent variable, while values less than 1 indicate a negative association (Hosmer et al., 2013). To evaluate the quality of the model, we examined the Akaike Information Criterion (AIC), Bayesian Information Criterion (BIC), the likelihood ratio chi-square test, and the area under the receiver operating characteristic curve (AUC). The final model yielded an AUC of 0.697, which suggests that the model correctly distinguishes between STEM career aspirants and non-STEM career aspirants about 70% of the time when comparing a random pair of cases (Fawcett, 2006).

Limitations

This study has several limitations that should be considered when interpreting the findings. First, the analysis relied on secondary survey data originally designed to assess college students' academic and personal development. As such, the available measures do not fully capture all six forms of CCW as defined by Yosso (2005). Second, we used aggregated racial/ethnic categories and applied effect coding to examine group-level differences in STEM career aspirations relative to the grand mean of the sample, rather than selecting a single racial group as the reference category. While this strategy improves model stability and avoids deficit-oriented comparisons, it also limits our ability to explore within-group heterogeneity, particularly among multiracial students. For instance, the lived experiences of students identifying as White-Asian may differ substantially from those identifying as Black-Latina/o/x, but such differences could not be captured in our models. Third, HERI's items measuring faculty mentorship frequency did not specifically assess mentorship activities related to supporting students' career aspirations; instead, most items in the faculty mentorship scale focus on support for graduate education preparation rather than career development, which constrained our ability to further explore how a broader range of mentorship activities might relate to students' career aspirations. Fourth, our findings are not generalizable, as institutions must pay to participate in HERI surveys; our sample does not represent the general population of women of color in STEM. Finally, although our sample includes students nested within 42 institutions, we applied a single-level model due to minimal between-institution variance. However, we acknowledge that institutional contexts play a critical role in shaping students' STEM career trajectories (Park et al., 2022).

Authors' Positionalities

All authors come from diverse backgrounds and are committed to recognizing and amplifying the unique strengths and assets of historically URGs. As equity-focused researchers, we approached this data-driven study through an anti-deficit and transformative perspective. The first author is a first-generation Asian woman doctoral student whose research focuses on the STEM trajectories and science identities of WOC and other underrepresented students. The second author identifies as a white, gay, cisgender male faculty member. The third author is a Latin American male with a background in STEM and is currently a doctoral student in Higher Education. His research explores how institutional environments shape the college experiences and trajectories of students facing unique challenges both in the U.S. and internationally. The fourth author is an Asian male who majored in STEM and is a doctoral candidate with interests focused on information literacy and information justice in higher education. Collectively, our positionalities have guided us to critically examine systemic barriers in STEM education, highlight the strengths of underrepresented groups, and advocate for equitable and inclusive practices in higher education.

Findings

Exploring Background Factors Linked to Sustained STEM Career Aspirations

To answer research question 1, we examined whether background characteristics differed significantly between WOC who maintained STEM career aspirations and those who shifted to non-STEM career aspirations among WOC who entered college with initial STEM career aspirations. Table 2 presents the comparison results. Significant differences emerged across freshman year major fields ($\chi^2 = 27.7$, $p < 0.001$). WOC majoring in Math and Computer Science (76.0%) and Engineering (73.8%) in their freshman year were more likely to report sustained interest in pursuing a STEM career by the end of their senior year, while those in non-STEM majors were more likely to shift their career aspirations to non-STEM fields (47.4%). The disparities are evident in students' academic achievement. A higher percentage of WOC who maintained STEM career pursuits in their senior year had relatively higher college GPAs ($\chi^2 = 30.9$, $p < 0.001$). For example, 78.4% of WOC with an A or A+ GPA reported consistent STEM career aspirations in their senior year, compared to 54.3% of those with a C+ average, indicating a potential association between college GPAs and persistence of STEM career aspirations. The chi-square test also revealed that Asian (71.4%) and Multiracial (71.3%) students were more likely to maintain STEM career aspirations, while Hispanic students reported the highest rate of STEM career aspiration attrition.

We also observed differences based on first-generation status ($\chi^2 = 8.29$, $p < 0.01$). First-generation students (61.3%) demonstrated a relatively lower persistence rate for STEM career aspirations compared to non-first-generation peers (70.3%). Students who had at least one parent working in STEM fields (72.6%) reported a higher likelihood of maintaining STEM career aspirations, compared to those whose parents did not work in STEM fields. In contrast, other demographic characteristics, such as citizenship status, did not show significant differences between WOC with and without STEM career aspirations.

The Role of CCW in Predicting STEM Career Aspirations

To answer research question 2, we estimated a nested logistic regression model to examine how background characteristics and four different forms of CCW associate the likelihood of maintaining STEM career aspirations among senior WOC who entered college with initial STEM career aspirations. Table 3 displays results across two models. Model 1 included demographic and background variables, along with WOC students' initial aspirational and resistant capital they brought with them upon college entry from the TFS. Model 2 incorporated indicators of navigational and social capital they developed during college from the CSS. For these two models, we report the regression coefficient (B), odds ratio (OR), standard error (SE), and p-value. Model fit improved at each step. The AUC increased from 0.651 in Model 1 to 0.697 in Model 2. Both AIC and BIC decreased from Model 1 to Model 2, indicating that the inclusion of variables from the CSS leads to a better-fitting model predicting senior-year STEM career aspirations while avoiding overfitting. Likelihood ratio tests confirmed that incorporating navigational and social capital measures from the CSS significantly improved model performance (LRT $\chi^2 = 58.40$, $p < 0.001$).

Among background characteristics, college GPA emerged as a statistically significant predictor of STEM career aspiration persistence. Specifically, in Model 2, each one standard deviation increase in GPA was associated with a 24% higher likelihood of reporting a STEM career aspiration in their senior year (OR = 1.24, $p < 0.001$). Additionally, students' initial major fields significantly related to their likelihood of sustaining their STEM career intentions by their senior year. Students who entered college in non-STEM majors were significantly less likely to maintain STEM career aspirations compared to those in Biological and Life Sciences (OR = 0.64, $p < 0.05$). By contrast, students who majored in Engineering (OR = 1.46, $p < 0.05$) and Math and Computer Science (OR = 1.80, $p < 0.05$) in their freshmen year emerged as significantly more likely to continue to aspire toward STEM careers, compared to who majored in Biological and Life Sciences. Other demographic and institutional variables, including race/ethnicity, citizenship status, first-generation status, parental STEM occupation, institutional type, and institutional selectivity, were not significantly associated with variation in students' STEM career aspiration persistence in the final model.

Among the aspirational and resistant capital measures, WOC who reported a lifelong goal of wanting to make a scientific contribution had a significantly higher likelihood of maintaining STEM career aspirations (OR = 1.20, $p < 0.01$). However, this indicator became non-significant after accounting for navigational and social capital in Model 2. In both models, students who reported a stronger commitment to community leadership and advocacy for social change, measured through their social agency, were significantly less likely to express intentions to maintain their initial STEM career aspirations (OR = 0.83, $p < 0.01$).

Finally, our results show that measures of navigational and social capital significantly predicted WOC students' likelihood of sustaining their STEM career intentions. WOC who participated in undergraduate research were 70% more likely than non-participants to maintain STEM career aspirations (OR = 1.70, $p < 0.001$). Similarly, students who joined campus programs supporting STEM careers, such as BUILD, MARC, HHMI, and S-STEM, showed a significantly increased likelihood of maintaining STEM career aspirations (OR = 1.71, $p < 0.001$). Interestingly, results showed that WOC who participated in internship programs during college were less likely to maintain STEM career aspirations (OR = 0.63, $p < 0.001$), suggesting that these internships may have been situated in industries with weaker ties to STEM-related pathways. Regarding social capital, students who engaged in pre-professional or departmental clubs were significantly associated with higher persistence rates (OR = 1.45, $p < 0.01$), while WOC who had frequent faculty mentorship and advisor support for career planning did not show significant differences in STEM career aspiration maintenance.

Discussion

While a growing body of literature has explored how CCW influences URGs students' STEM interests, science identity, and STEM trajectories (Hiramori et al., 2024; Lawson & Fong, 2024; Nhien, 2024; Revelo & Baber, 2018; Sablan, 2019), this study contributes to the discourse by centering undergraduate WOC and critically reflecting on the influences of CCW on WOC's persistence of STEM career aspirations. In reporting the key influences, WOC who expressed a strong desire to make scientific contributions had better odds of sustaining the STEM career aspirations during college. Conversely, students who entered college with a stronger commitment to social change and community leadership were less likely to intend pursue STEM careers by the end of college, suggesting tension between motivations for social justice and STEM career pathways. Notably, findings related to navigational and social capital, such as participation in undergraduate research and targeted STEM support programs (BUILD, MARC, HHMI, and S-STEM), echo previous research which has identified these as critical supports for URG students in STEM (Hernandez et al., 2018; Wilson et al., 2018), and were significantly associated with STEM career aspiration persistence in this study. Our findings call for more critical examination of how different forms of CCW contribute to maintaining WOC STEM career aspirations.

Background Characteristics of WOC Maintaining STEM Career Aspirations

The higher persistence rates in Math, Computer Science, and Engineering may stem from early science identity formation (Espinosa, 2011), clear connections between coursework and industry applications (Wang & Degol, 2017), and visible market demand in rapidly expanding technology fields (NCSES, 2023). By contrast, students who entered college with non-STEM majors but initially aspired to STEM careers may face greater challenges in developing the career motivations, science identities, and knowledge and skills foundations necessary for STEM career persistence (Barbara, 2021). Our analyses also revealed that academic achievement, as measured by college GPA, significantly influences STEM career aspiration persistence, with higher-achieving students demonstrating a greater likelihood of maintaining their initial STEM career goals. This finding aligns with previous research suggesting that academic success reinforces students' commitment to STEM pathways (Rosenzweig et al., 2024; Xu, 2016).

Community Cultural Wealth and the Persistence of STEM Career Aspirations

Our findings highlight aspirational capital as a significant and positive predictor of WOC students' STEM career aspirations. Aspirational capital, as the ability to maintain hopes and dreams for the future despite structural barriers (Yosso, 2005), was evident in students who expressed a strong desire to make scientific contributions. These students were significantly more likely to aspire to STEM-related careers by their senior year. These intrinsic goals may help WOC to sustain their interest in pursuing STEM-related careers as they navigate their undergraduate curricula (Fredricks et al., 2025). This finding is consistent with prior scholarship emphasizing the motivational power of scientific purpose and social impact in shaping students' long-term goals (Connors, 2023; Sax et al., 2018).

Similar to prior research (see Denton et al., 2020), our findings indicate that both navigational and social capital served as significant predictors that may influence WOC's STEM career aspiration persistence. As previous studies have underscored the importance of "high-impact practices" for URGs, such as undergraduate research, faculty mentorship, and formal and informal STEM program or group engagement (Chang et al., 2014; Cuellar & Gonzalez, 2021; Rincón & Rodriguez, 2021), our study also demonstrated that undergraduate research opportunities, formal programs supporting STEM careers (e.g., BUILD, MARC, HHMI, S-STEM), and informal pre-professional STEM learning clubs have a salient, positive association with WOC's likelihood to intend to pursue STEM careers by the end of their senior year in college.

Our research findings regarding participation in internship programs also echo previous studies that demonstrated WOC who have experienced working in STEM fields may choose to leave STEM career trajectories (Garibay et al., 2013). This may be due to STEM work cultures that cater to men and dominant groups, resulting in WOC shifting their career aspirations and directions (Rhoton, 2011). This finding may also reflect that students in our sample had already decided upon a different, non-STEM career path and chose an internship aligned with that pathway. Additionally, faculty mentorship did not show any significant association. This may be due to our frequency-based faculty mentorship measurement approach, as the scale primarily captures academic support rather than career-oriented mentoring that might more directly influence students' career aspirations.

Finally, regarding resistant capital, as prior research indicates, women and students from URGs are more likely than their majority-status peers to view STEM as a tool for promoting social change (Sax et al., 2017). Similarly, researchers found that WOC who had altruistic science identities often valued science as a means to give back to their communities and support others, which contrasted with some of the dominant norms of the STEM field (Carlone & Johnson, 2007; Garibay, 2018). However, this study found that WOC who expressed strong commitments to community leadership and social advocacy were less likely to pursue STEM careers by their senior year. These findings may suggest that many STEM subfields are disconnected from broader social impact (Garibay, 2018). Students who are driven by personal values centered on helping others and contributing to their community may be discouraged from pursuing STEM majors if they perceive these fields as lacking alignment with prosocial or communal goals (Brown et al., 2015). Those with strong commitments to activism and social justice may be more likely to pursue careers in fields such as social sciences and humanities, where opportunities to engage in systemic change are perceived as more direct and attainable.

Implications for Future Studies and Practice

Throughout our research process, we have engaged in critical reflection regarding our model choice and the extent to which they aligned with our CCW conceptual framework. This includes decisions about the sample population, selection of CCW measures and data collection time points, our approaches for addressing missing data and coding racial groups, and interpretation of our findings. We have aimed to describe these decision points in detail and explain how they reflected our equity-focused analytic perspective. While this study expands quantitative CCW research by exploring WOC's persistence in STEM career aspirations, it also highlights critical areas for future investigation.

We found that students who express stronger commitments to social change and community leadership are less likely to maintain their aspirations for STEM careers, suggesting that WOC who prioritize equity and advocacy may view STEM occupations as misaligned with those values. While there are increasing studies that highlight the importance of developing students' socially responsible STEM outcomes (Garibay, 2018), practices or strategies to retain these justice-oriented students on STEM trajectories remain unclear. Future research could investigate how justice-oriented WOC interpret STEM cultures and whether institutional practices could better support their community leadership, advocacy for social justice, and commitment to public welfare within STEM contexts. Additionally, we suggest future research should expand the conceptualization of resistant capital to include forms that originate from personal resistance experiences. For instance, students of color may resist negative messages about their abilities and career trajectories, transforming these experiences into motivation for an advanced degree or career pursuit (Espino, 2014). Future studies could investigate how WOC's resistance to negative college experiences might nourish and sustain their STEM career aspirations.

Moreover, rather than treating institutions as neutral backdrops, future research should consider incorporating multilevel modeling to examine how institutional environments actively shape the development of CCW and its influence on WOC students' STEM career aspirations persistence. This ecological perspective would allow scholars to explore how institutional factors, such as racial climate, the representation of WOC faculty, and the availability of justice-oriented STEM programs, influence the relationship between CCW and the persistence of STEM career aspirations. Such research would provide a more comprehensive understanding of how structural and individual factors interact to influence WOC's educational and career trajectories in STEM fields.

Finally, our findings highlight the value of embedding research opportunities, academic support, and inclusive extracurricular engagement into the experience of WOC in STEM. At a moment when federal support for such programs has waned, campus leaders have a responsibility to act boldly by taking steps to sustain and institutionalize these high-impact practices within the curriculum. These opportunities do more than merely provide students with technical skills, they also affirm students' potential, cultivate a sense of belonging, and offer tangible insights into what a future in STEM can look like. By centering the strengths and aspirations of WOC, colleges and universities can play a critical role in expanding access to meaningful and productive STEM career pathways.

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Table 1. Descriptive statistics for the study samples (N=1,460)

	%	mean (SD)
Dependent Variable		
Senior Year STEM Career Aspiration	68.6%	
Background Characteristics		
Race/Ethnicity		
Asian	42.8%	
Black	10.3%	
Hispanic	18.1%	
Multiracial	27.0%	
Other	1.8%	
Freshman Year Major Fields		
Non-STEM	11.8%	
Biological & Life Sciences	40.7%	
Engineering	21.9%	
Health Professions	14.4%	
Math & Computer Science	7.1%	
Physical Science	4.0%	
Citizenship Status		
International students	4.7%	
Permanent residents	4.2%	
U.S. citizens	89.1%	
None of the above	1.9%	
First-generation	19.5%	
One of parents works in STEM	33.8%	
College GPA		5.94 (1.52)
Institutional Context		
Institution Selectivity		1220 (154)
Institutional Type		
Public Institution	33.5%	
Private Institution	66.5%	
CCW		
CCW - Aspirational Capital (TFS)		
Aspire to make a scientific contribution		2.46 (0.99)
Aspire to become an authority in the field		2.74 (0.87)
Aspire to gain colleagues recognition		2.77 (0.85)
CCW - Resistant Capital (TFS)		
Commitment to community leadership and advocacy for social change		50.3 (10.2)
Motivated in civic, electoral, and political activities		50.6 (10.3)
Plan to participate in student protests or demonstrations		2.21 (0.84)
CCW - Navigational Capital (CSS)		
Participated in undergraduate research opportunities	45.0%	
Joined campus programs supporting STEM careers (e.g., BUILD, MARC, HHMI, S-STEM)	31.6%	
Participated in an internship program	59.6%	
CCW - Social Capital (CSS)		
Frequency of faculty mentorship		55.7 (12.9)
Advisor support for career planning		2.2 (0.65)
Engaged in pre-professional or departmental clubs	47.3%	

Table 2. Cross-tabulation and Chi-Square Comparisons of Senior Year STEM vs. Non-STEM Career Aspirations (N=1,460)

	STEM Career Aspiration (N=1001)	Non-STEM Career Aspiration (N=459)	Pearson χ^2
Race/Ethnicity			$\chi^2(4) = 15.8$**
Asian	71.4% (446)	28.6% (179)	
Black	66.7% (100)	33.3% (50)	
Hispanic	58.7% (155)	41.3% (109)	
Multiracial	71.3% (281)	28.7% (113)	
Other	70.4% (19)	29.6% (8)	
Freshman Year Major Fields			$\chi^2(5) = 27.7$***
Non-STEM	52.6% (91)	47.4% (82)	
Biological & Life Sciences	69.0% (410)	31.0% (184)	
Engineering	73.8% (236)	26.2% (84)	
Health Professions	67.6% (142)	32.4% (68)	
Math & Computer Science	76.0% (79)	24.0% (25)	
Physical Science	72.9% (43)	27.1% (16)	
College GPA			$\chi^2(7) = 30.9$***
A or A+	78.4% (203)	21.6% (56)	
A-	73.9% (235)	26.1% (83)	
B+	67.0% (219)	33.0% (108)	
B	63.3% (207)	36.7% (120)	
B-	64.0% (80)	36.0% (45)	
C+	54.3% (38)	45.7% (32)	
C	54.8% (17)	45.2% (14)	
D	66.7% (2)	33.3% (1)	
Citizenship Status			$\chi^2(3) = 3.49$
International students	68.1% (47)	31.9% (22)	
Permanent residents	79.0% (49)	21.0% (13)	
U.S. citizens	68.2% (887)	31.8% (414)	
None of the above	64.3% (18)	35.7% (10)	
First Generation Status			$\chi^2(1) = 8.29$**
First-generation	61.3% (174)	38.7% (110)	
Not first-generation	70.3% (827)	29.7% (349)	
Parental Career			$\chi^2(1) = 5.40$*
At least one parent works in STEM	72.6% (358)	27.4% (135)	
Neither parent works in STEMH	66.5% (643)	33.5% (324)	

Table 3. Results of logistic regression on WOC's Senior Year STEM career aspirations

	Model 1					Model 2				
	B	OR	SE	p-value		B	OR	SE	p-value	
Constant	0.91	2.48	0.45	0.045	*	0.79	2.20	0.49	0.103	
Background Characteristics										
Race/Ethnicity										
Asian	0.03	1.03	0.13	0.843		0.05	1.05	0.13	0.716	
Black	0.07	1.07	0.18	0.712		0.03	1.03	0.18	0.863	
Hispanic	-0.27	0.76	0.15	0.077		-0.20	0.82	0.16	0.194	
Multiracial	0.09	1.10	0.14	0.521		0.12	1.13	0.14	0.401	
Other	0.09	1.09	0.30	0.750		0.00	1.00	0.31	0.920	
Freshman Year STEM Major Fields (ref. Biological & Life Sciences)										
Non-STEM	-0.60	0.55	0.19	0.001	**	-0.45	0.64	0.20	0.021	*
Engineering	0.26	1.30	0.17	0.110		0.38	1.46	0.17	0.029	*
Health Professions	0.05	1.05	0.18	0.786		0.21	1.24	0.19	0.260	
Math & Computer Science	0.36	1.43	0.26	0.164		0.59	1.80	0.27	0.028	*
Physical Science	0.12	1.13	0.31	0.695		0.17	1.19	0.32	0.596	
Citizenship Status (ref. None of these)										
International students	-0.44	0.64	0.51	0.381		-0.75	0.47	0.53	0.158	
Permanent residents	0.28	1.32	0.53	0.603		0.02	1.02	0.55	0.970	
U.S. citizens	-0.12	0.89	0.43	0.784		-0.34	0.71	0.45	0.449	
First-generation (ref. non first generation)	-0.20	0.82	0.16	0.198		-0.25	0.78	0.16	0.123	
One of parents works in STEM (ref. none of)	0.11	1.11	0.13	0.419		0.08	1.08	0.13	0.561	
College GPA	0.28	1.33	0.05	0.000	***	0.21	1.24	0.06	0.000	***
Institutional Context										
Institution Selectivity	-0.01	0.99	0.07	0.867		0.01	1.01	0.07	0.836	
Private Institution (ref. Public)	0.01	1.01	0.16	0.937		0.04	1.04	0.16	0.816	
CCW										
CCW - Aspirational Capital (TFS)										
Aspire to make a scientific contribution	0.18	1.20	0.07	0.009	**	0.12	1.13	0.07	0.095	
Aspire to become an authority in the field	0.03	1.03	0.07	0.706		0.00	1.00	0.08	0.964	
Aspire to gain colleagues recognition	0.10	1.10	0.07	0.195		0.10	1.10	0.08	0.207	
CCW - Resistant Capital (TFS)										
Commitment to community leadership and advocacy for social change	-0.21	0.82	0.08	0.006	**	-0.19	0.83	0.08	0.013	*
Motivated in civic, electoral, and political activities	-0.12	0.89	0.07	0.072		-0.13	0.88	0.07	0.069	
Plan to participate in student protests or demonstrations	-0.01	0.99	0.06	0.835		-0.04	0.96	0.06	0.535	
CCW - Navigational Capital (CSS)										
Participated in undergraduate research opportunities						0.53	1.70	0.14	0.000	***
Joined campus programs supporting STEM careers (e.g., BUILD, MARC, HHMI, S-STEM)						0.54	1.71	0.15	0.000	***
Participated in an internship program						-0.46	0.63	0.13	0.000	***
CCW - Social Capital (CSS)										
Frequency of faculty mentorship						0.07	1.08	0.06	0.252	
Advisor support for career planning						-0.10	0.90	0.06	0.107	
Engaged in pre-professional or departmental clubs						0.37	1.45	0.13	0.003	**
Model Fitness										
AIC	1770.69					1724.29				
BIC	1897.56					1882.88				
AUC (ROC)	0.651					0.697				
LRT χ^2						58.40***				

Appendix A. STEM Career Classification in 2015 TFS and 2019 CSS

STEM Occupations	Natural Resource Specialist/Environmentalist, College Faculty, Computer Programmer, Developer, Computer Systems Analyst, Web Designer, Engineer, Research Scientist (e.g., Biologist, Chemist, Physicist), Scientific Researcher (TFS only), Statistician (TFS only), Dietician/Nutritionist, Home health worker, Medical/Dental Assistant (Hygienist, Lab Tech, Nursing Asst.), Registered Nurse, Therapist (e.g. physical, occupational, speech), Clinical Psychologist, Dentist/orthodontist, Medical doctor/surgeon, Optometrist Pharmacist, Veterinarian

Appendix B. STEM Major Classification in 2015 TFS

Biological & Life Sciences	Agriculture, Agriculture/Natural Resources, Animal Biology (zoology), Biochemistry/Biophysics, Biology (general), Botany, Ecology & Evolutionary Biology, Environmental Science, Marine (life) Science, Marine Biology, Microbiology, Microbiology or Bacteriology, Molecular Cellular & Developmental Biology, Neurobiology/Neuroscience, Other Biological Science, Plant Biology (botany), Zoology
Engineering	Aeronautical or Astronautical Eng, Aerospace/Aeronautical/Astronautical Engineering, Biological/Agricultural Engineering, Biomedical Engineering, Chemical Engineering, Civil Engineering, Computer Engineering, Electrical or Electronic Engineering, Electrical/Electronic Communications Engineering, Engineering Science/Engineering Physics, Environmental/Environmental Health Engineering, Industrial Engineering, Industrial/Manufacturing Engineering, Materials Engineering, Mechanical Engineering, Other Engineering
Math & Computer Science	Computer Science, Data Processing or Computer Programming, Mathematics, Mathematics/Statistics, Other Math and Computer Science, Statistics
Physical Science	Astronomy, Astronomy & Astrophysics, Atmospheric Sciences, Chemistry, Earth & Planetary Sciences, Earth Science, Marine Sciences, Other Physical Science, Physics
Health Professions	Clinical Laboratory Science, Health Care Administration/Studies, Health Technology, Medical Dental Veterinary, Nursing, Other Health Profession, Other Professional, Pharmacy, Therapy (occupational physical speech)
Non-STEM	Accounting, Anthropology, Architecture/Urban Planning, Art fine and applied, Building Trades, Business Administration (general), Business Education, Classical and Modern Language and Literature, Communications (radio TV etc.), Computer/Management Information Systems, Criminal Justice, Drafting or Design, Economics, Electronics, Elementary Education, English (language & literature), Entrepreneurship, Ethnic Studies, Ethnic/Cultural Studies, Finance, Forestry, Geography, History, Home Economics, Hospitality/Tourism, Human Resource Management, International Business, Journalism, Journalism/Communication, Language and Literature (except English), Law Enforcement, Library or Archival Science, Library Science, Management, Marketing, Mechanics, Media/Film Studies, Military Science, Military Sciences/Technology/operations, Music, Music/Art Education, Other Arts and Humanities, Other Business, Other Education, Other Social Sciences, Other Technical, Philosophy, Physical Education/Recreation, Political Science (govt. international relations), Psychology, Public Policy, Real Estate, Secondary Education, Secretarial Studies, Security & Protective Services, Social Work, Sociology, Special Education, Speech, Speech or Theater, Theater/Drama, Theology/Religion, Women's Studies, Women's/Gender Studies

Appendix C. Item Parameters for the HERI Construct Scores Using Item Response Theory

TFS - Social Agency Score: Please indicate the importance to you personally of each of the following (1-4 scale)				
Item	A	C1	C2	C3
Influencing social values	1.78	2.93	-0.39	-3.56
Helping others who are in difficulty	1.71	5.78	1.77	-1.72
Participating in a community action program	2.86	2.8	-1.59	-5.13
Helping to promote racial understanding	2.77	3.04	-1.38	-4.98
Keeping up to date with political affairs	2.15	3.12	-0.85	-4.39
Becoming a community leader	2.65	2.58	-1.41	-4.96
TFS - Civic Engagement Score: Measures the extent to which students are motivated and involved in civic, electoral, and political activities (1-3, 1-4 scale)				
Item	A	B1	B2	B3
Demonstrated for a cause (e.g., boycott, rally, protest)	1.46	0.93	2.80	
Performed volunteer work	0.80	-2.82	0.73	
Helped raise money for a cause or campaign	1.42	2.09	3.38	
Publicly communicated my opinion about a cause (e.g., blog, email, petition)	1.11	-0.31	1.92	
Influencing social values	0.97	-1.96	0.37	2.42
Keeping up to date with political affairs	0.86	-1.50	0.74	2.67
CSS - Faculty Mentorship Score: How often have professors at your college provided you with (1-3 scale)				
Item	A	B1	B2	
Help in achieving your professional goals	3.85	-1.2	1.27	
Advice and guidance about your educational program	3.33	-1.44	1.25	
Emotional support and encouragement	2.64	-1.14	1.47	
Feedback on your academic work (outside of grades)	2.96	-1.29	1.28	
Encouragement to discuss coursework outside of class	3.04	-1.74	1.21	
Encouragement to pursue graduate/professional study	2.47	-1.41	1.28	
Help to improve your study skills	2.27	-1.04	1.68	
A letter of recommendation	1.93	-1.14	1.62	
An opportunity to work on a research project	1.31	-0.8	2.36	